Spatial Enhancement

Spatial Transform Filtering

Spatial Enhancement

- **Spatial enhancement modifies pixel values based on the surrounding pixel values**
- **Spatial enhancement deals largely with spatial frequency**
- **Spatial Frequency:** Refers to the "roughness" of the tonal variations occurring in an image, defined as
    - **the number of changes in brightness value per unit distance for any part of an image**

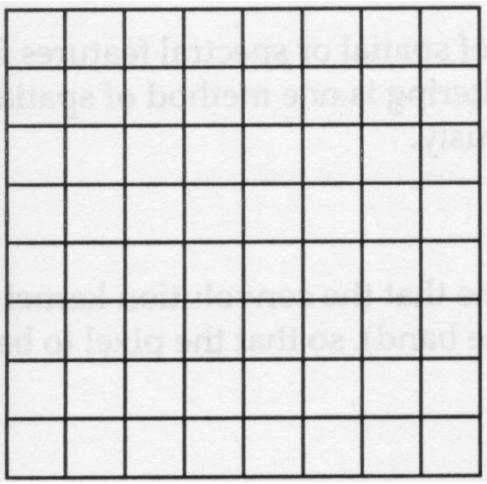
Spatial Frequency

- Any image can be seen as made of two basic components of information:
  - a high spatial frequency
  - a low frequency component
    - their sum makes up the original image

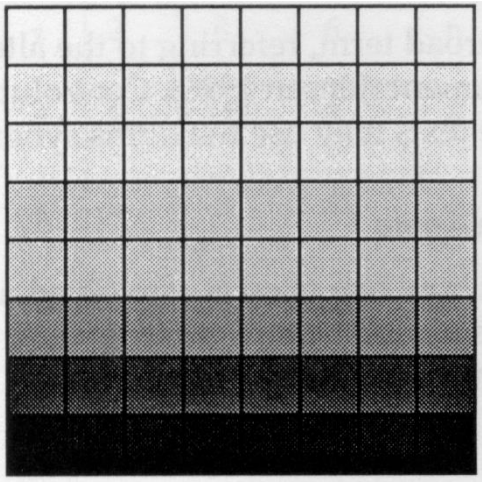
Spatial Frequency

- Low spatial frequency: If the gray scale in an image change only gradually over a large number of image pixels
- High spatial frequency: Large brightness change over a very few pixels in the image (can be thought of as a change)

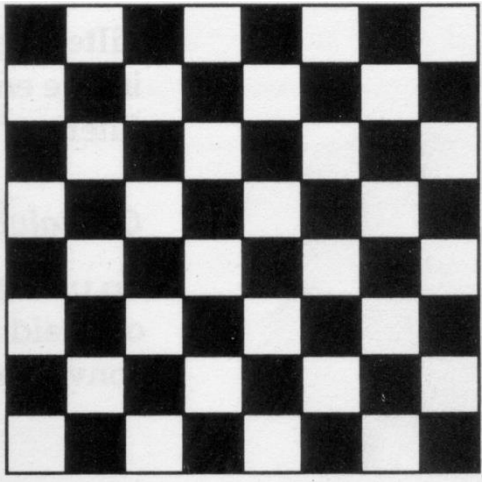
Zero Spatial Frequency



Low Spatial Frequency



High Spatial Frequency



Spatial Frequency

- **High frequencies** describe sudden\big changes in pixel values along a short distance, this is an area of high frequency details (linear features:, rivers, roads, pipelines, the borders of area features, buildings, etc)
- **Low frequencies** describe gradual\small changes in pixel values in an image (such as: water bodies, agricultural fields, natural vegetation, etc)

Spatial Enhancement

- Human interpreter will quickly be able to see things with high spatial frequency and high contrast
- Spatial enhancement is to provide techniques that will emphasize these areas of high frequency so they are easier to pick up by the human eye
- In image areas showing high spatial frequencies, there are generally edges (low and high brightness values
- Examples of the Spatial enhancement techniques are often called "edge enhancement" or "edge detection"

Spatial Filtering

- Spatial frequency in remotely sensed imagery may be enhanced or repressed using Spatial convolution filtering primarily based on the use of convolution masks

Spatial Filtering

- **Filtering: An operation that changes the value of a pixel according to the "gray" values of its surrounding pixels**
- **This process of creating output values that are some function of an input neighborhood is called convolution filtering**
- **Convolution filtering is a method of spatial filtering**
- **Filters operate on the image data in the row, column space, the spatial domain**

Filters

- **Spectral filters**: which serve to block or pass energy over various spectral ranges
- **Spatial filters** emphasize or deemphasize image data of various spatial frequencies
- **Spatial filters**
  - **High Pass Filter**
  - **Low Pass Filter**
  - **Edge detectors**

Basic Principle of a Filter

- Although there are many kinds of filters that can be applied on an image, the basic principle of their operation are similar:
  - Window size of the filter is to be defined, usually having an odd number of columns and rows (the number of the rows equals that of the columns)
  - Also known as kernel (same window size)
  - A weight is assigned to each pixel in the window

Convolution kernel

- The numbers in the matrix are often called coefficients(ci), because they are used as such in the mathematical equations
- **Mask template =**

C 1	C 2	C 3
C 4	C 5	C 6
C 7	C 8	C 9

  - $c_1 \times BV_1$ $c_2 \times BV_2$ $c_3 \times BV_3$

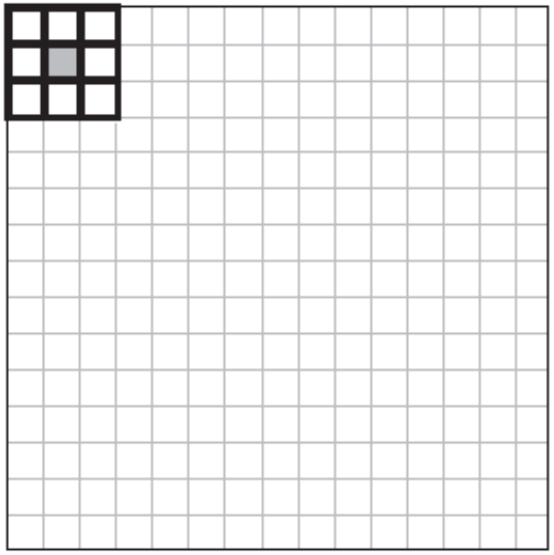
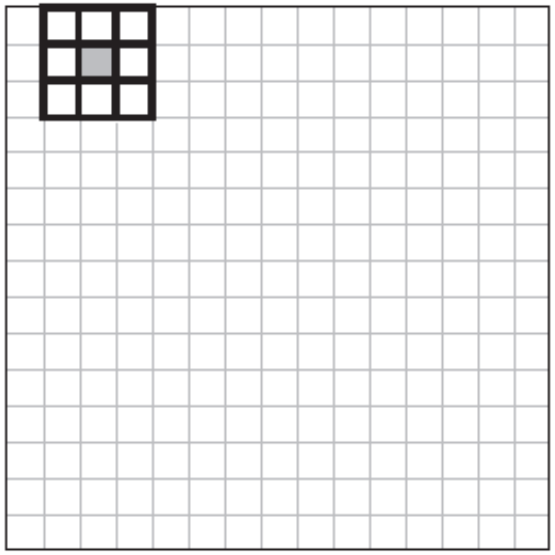
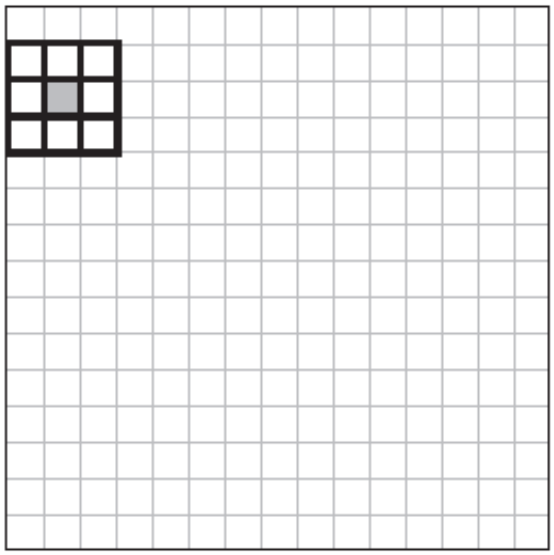
- **Mask template =** $c_4 \times BV_4$ $c_5 \times BV_5$ $c_6 \times BV_6$
  $c_7 \times BV_7$ $c_8 \times BV_8$ $c_9 \times BV_9$

- **The primary input pixel under investigation at any one time is always BV_5 and accordingly, its value is changed in the output image**
- **Mask size determines the degree of smoothing and loss of details**

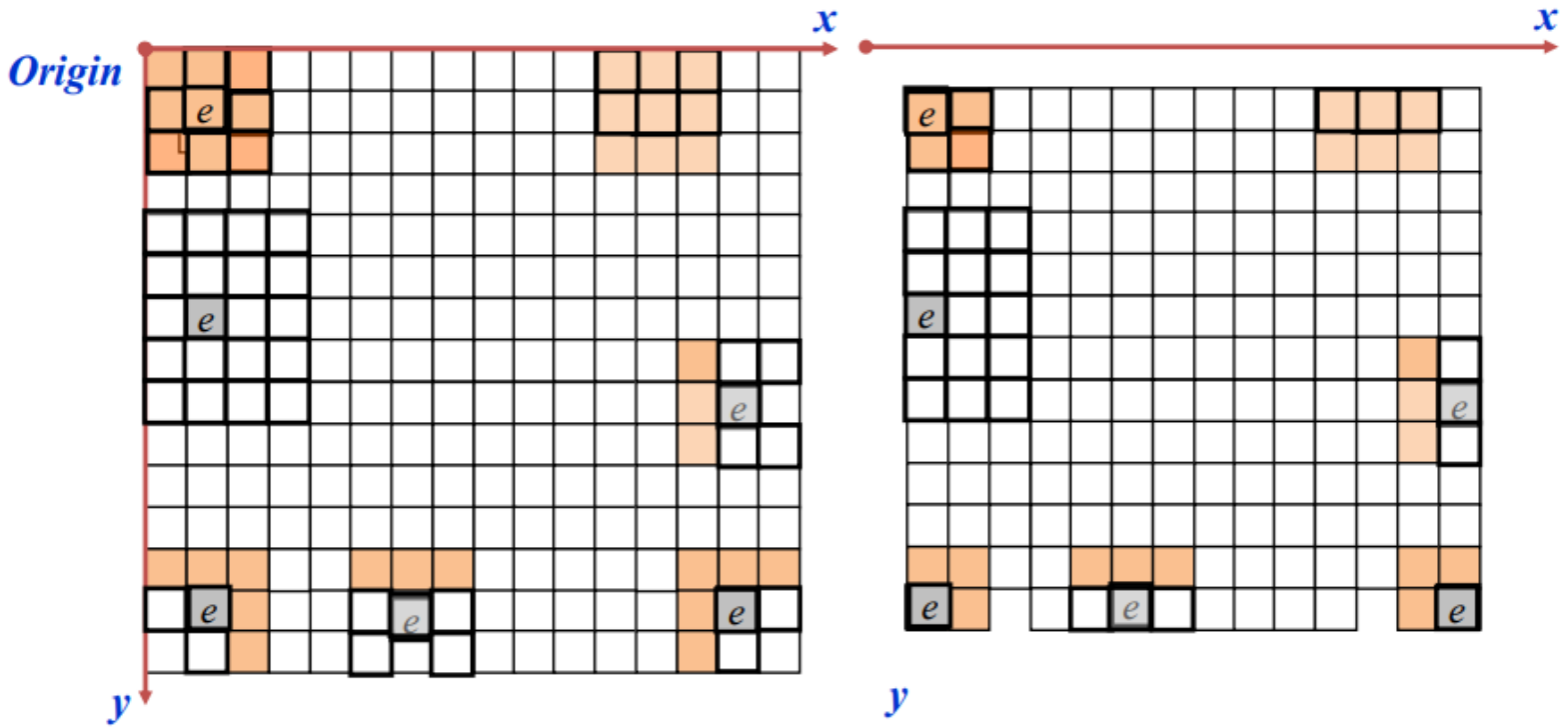
Basic Principles of a Filter

- Moving the window over the whole image, this operation is preformed on a pixel by pixel basis
- The output value at each output pixel is equal to the sum of the individual products of each filter coefficient with the corresponding pixel value
- Combination of SHIFT, MULTIPLICATION and ADDITION

Convolution Filters

Moving a Filter



Convolution kernel

- **Filter Coefficients:**
	c1	c2	c3
Mask template =	c4	c5	c6
	c7	c8	c9

- **Coefficients**
  - Positive: Blurs the image
  - Alternatively positive and negative : Sharpen the image
  - Sum to 1: Brightness retained
  - Sum to zero: Dark image

Convolution

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

(*a*) Kernel

67	67	72
70	68	71
72	71	72

(*b*) Original image DNs

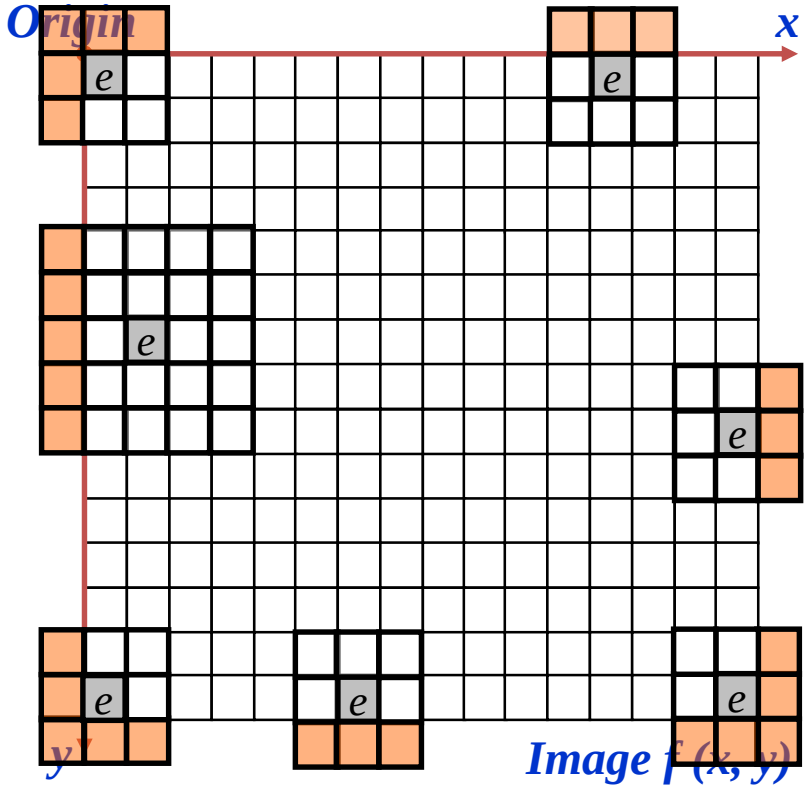
	70	

(*c*) Convolved image DN

Convolution: 1/9(67) + 1/9(67) + 1/9(72) + 1/9(70) + 1/9(68) + 1/9(71) + 1/9(72) + 1/9(71) + 1/9(72) = 630/9 = 70

The Edges!

At the edges of an image, pixels are missing to form a neighbourhood



- Pad the image
  - Typically with either all white or all black pixels
- Replicate border pixels
- Truncate the image

Applying Filter

In order that pixels along the perimeter of an image can be a center for the window, the perimeter pixels are duplicated (temporarily) during the filter operation

Original image

12	18	4	0
11	2	12	5
3	23	1	8

With duplicated edges to perform a 3*3 filter

12	12	18	4	0	0
12	12	18	4	0	0
11	11	2	12	5	5
3	3	23	1	8	8
3	3	23	1	8	8

The 3*3 filter window

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Output filtered image

12	10.3	7	3.3
10.6	9.6	8.1	4.8
9.1	8.8	9.2	6.2

12x1/9+ 18x1/9+ 4x1/9+11x1/9+2x1/9+12x1/9+ 3x1/9+23x1/9+1x1/9= 9.6

Filters

- **Low-pass filters**
  - Emphasizes low spatial frequencies to reduced / eliminate noise within an image
  - Generate an image that will appear smooth or blurred relative to the original
  - Useful for increasing the homogeneity of images that have too many speckles or for smoothing stair steps
  - Removing “salt” and “pepper” noise

High Pass Filters

- **Emphasize high spatial frequency (to emphasize border or edges, changes at borders or edges)**
- **High pass filters will enhance contrast differences at edges, thus sharpening linear features**
- **If a road exists with a definite contrast difference to its surroundings, the edge enhancement filter should emphasize those differences and make the road more distinct**

Low & High Pass filtering Convolution Mask

Low Pass Convolution Mask =

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

High Pass Convolution Mask =

-1	-1	-1
-1	8	-1
-1	-1	-1

Low Pass Filters

- MEAN and GAUSSIAN are used to smooth an image
- In MEAN filter all pixels have an equal weight
- While the GAUSSIAN, “distance weighted” the far a pixel is located from the central pixel, its weight is smaller

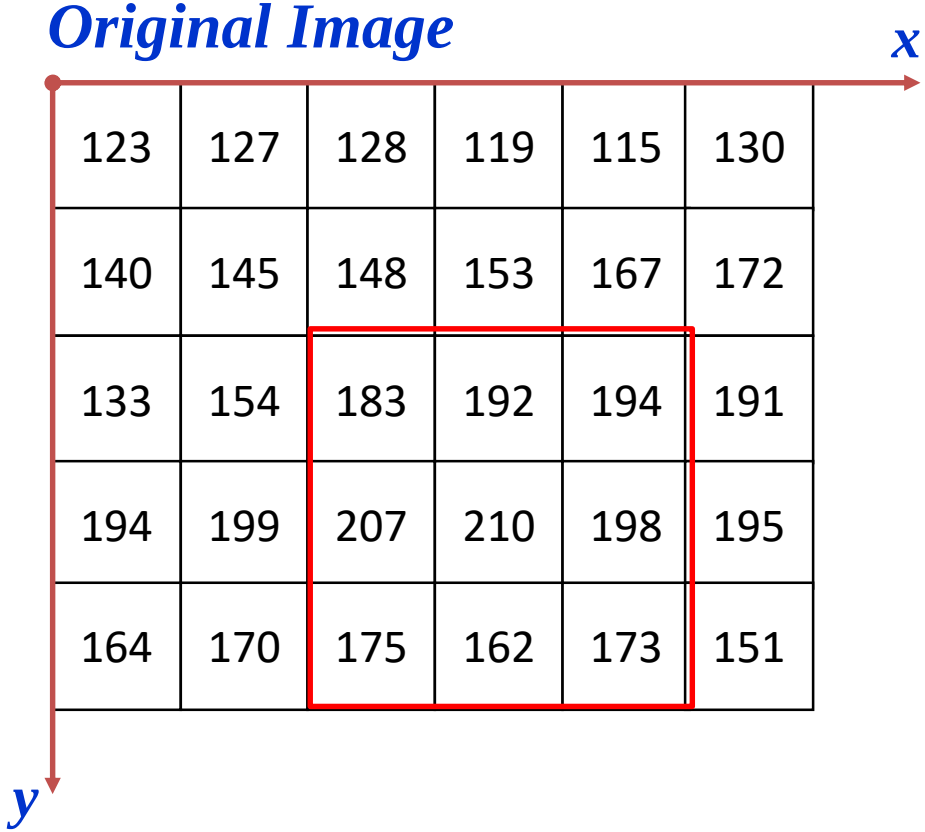
Low Pass Filters

- **MEAN and Weighted MEAN**
  - Replaces each pixel with an average of all the values in its neighbourhood
  - Assumes pixels to be like their neighbours
  - Used to smooth an image
  - In the MEAN filter, all pixels have an equal weight
  - While the “distance weighted”
    - if want the closest pixels to have a higher influence on the output

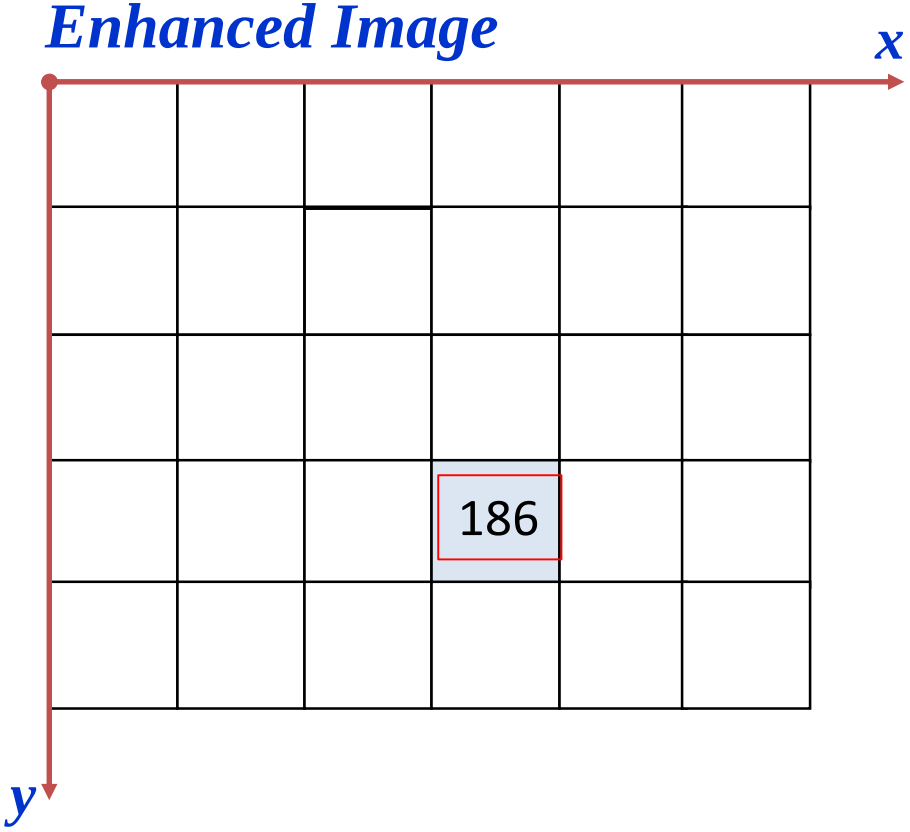
Simple Neighbourhood Operations

Mean

Original Image



Enhanced Image



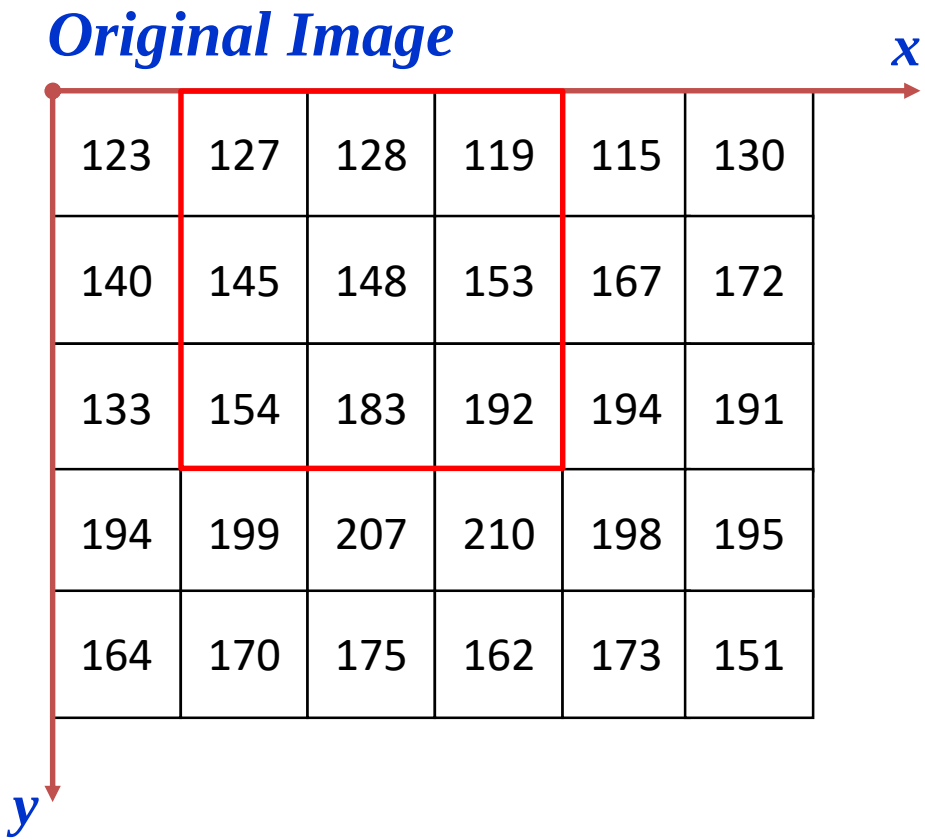
Simple Filters (Neighbourhood Operations)

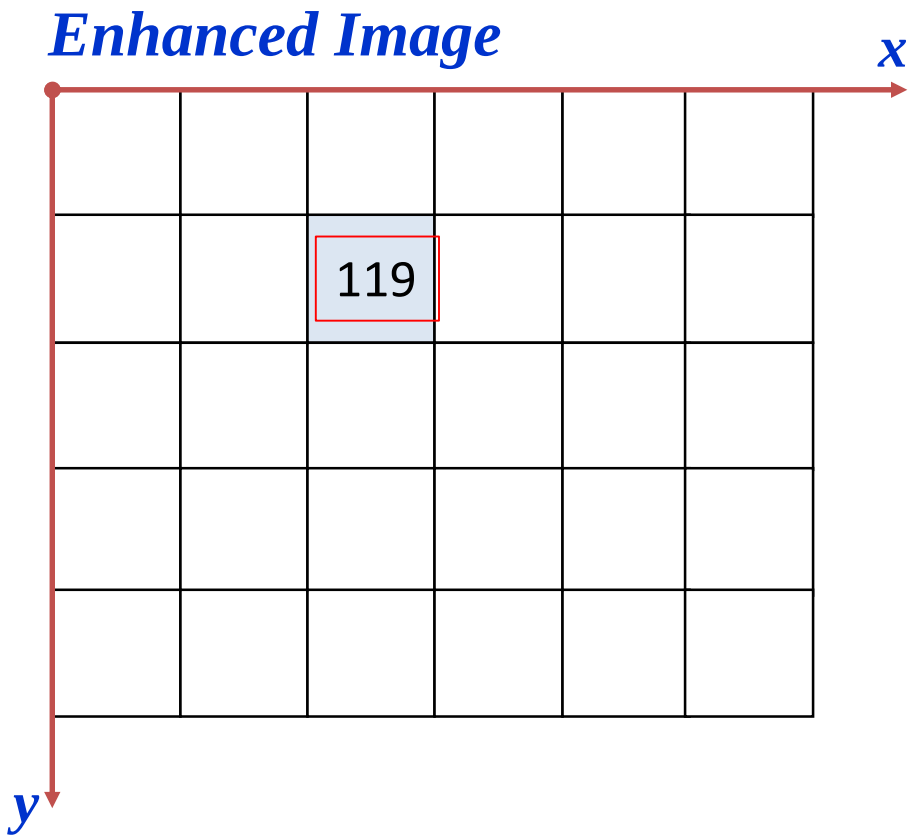
- Some simple neighbourhood operations include:
  - **Mean:** Averaging to smooth an image
  - **Min:** Set the pixel value to the minimum in the neighbourhood
  - **Max:** Set the pixel value to the maximum in the neighbourhood
    - Useful for visual analysis only
  - **Median:** The median value of a set of numbers is the midpoint value in that set (e.g. from the set [1, 7, 15, 18, 24] 15 is the median)
    - Used to minimize random noise

Simple Neighbourhood Operations

Min

Set the pixel value to the minimum in the neighbourhood

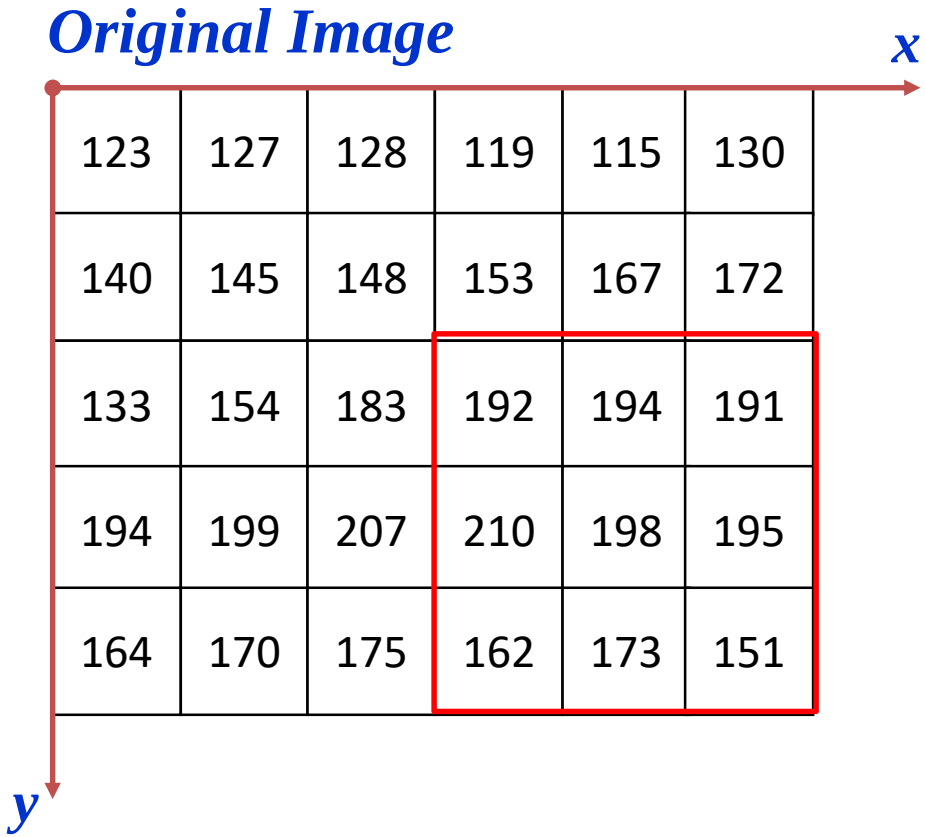


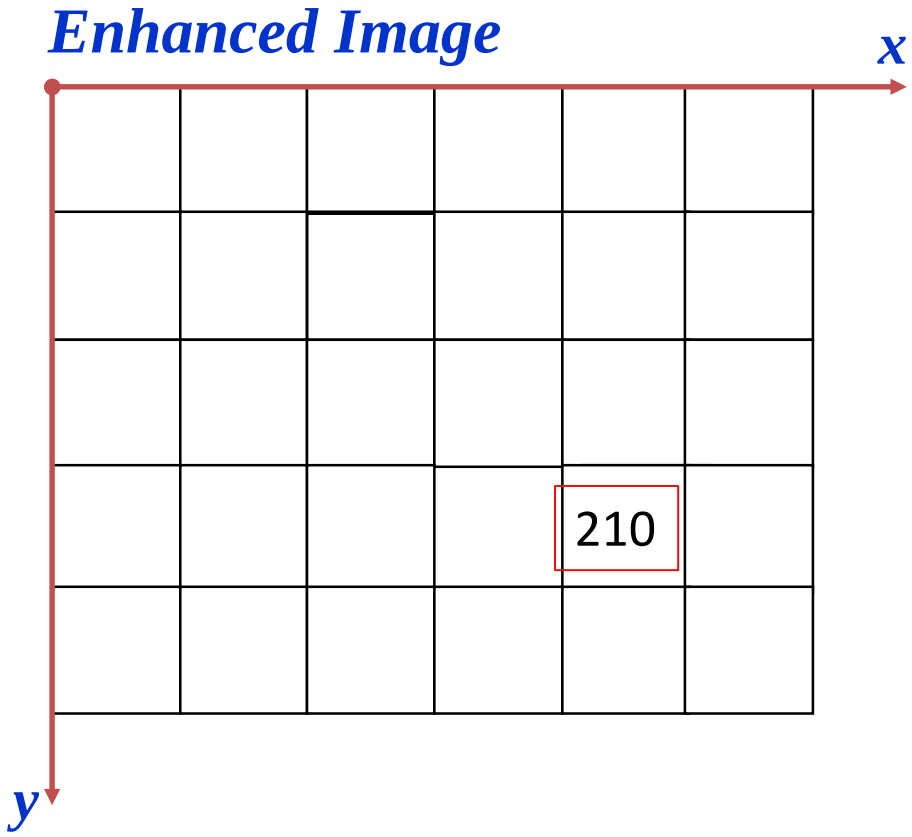


Simple Neighbourhood Operations

Max

Set the pixel value to the maximum in the neighbourhood





Simple Neighbourhood Operations

Median

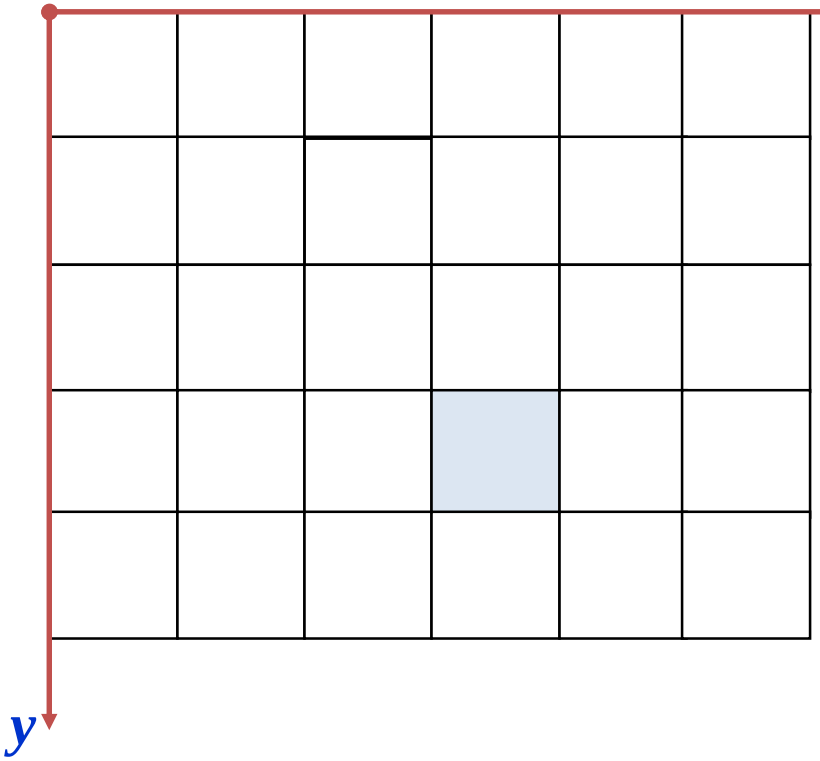
Original Image

123	127	128	119	115	130
140	145	148	153	167	172
133	154	183	192	194	191
194	199	207	210	198	195
164	170	175	162	173	151

x

y

Enhanced Image



x

y

162,173,175,183,192,194,198,207,210

?